transparency of any ML insights must remain explainable and optional.

Multi-language support: Accessibility is another area we're focusing on. Right now, the UI is English-only, but we're planning to add multilingual support using language files and client-side translation libraries. This would allow clinics or wellness programmes in different regions to adapt the interface for local use.

Analytics dashboards: For doctors and admins, we'd like to add analytics panels that give them insight into:

- Group participation trends
- Appointment volumes over time
- Resource engagement (e.g., which documents are most read)

These metrics could help users make data-driven decisions without needing external tools.

Better mobile experience: Although the current frontend is responsive, we want to fine-tune the mobile UI so that frequent tasks like joining a group or scheduling an appointment feel more native. We're considering wrapping the app in a progressive web app (PWA) shell or using something like React Native if offline functionality becomes a requirement.

Building SereneCare has been more than just a coding exercise -- it's been a way for us to understand how technology can genuinely improve access to healthcare. From the start, our goal was to create a platform that didn't just 'work', but felt intuitive and secure for patients, doctors, and administrators alike.

By using a modular architecture, role-based workflows, and a modern open source stack, we were able to build something flexible enough to adapt but structured enough to be maintained long-term. Features like real-time updates, resource sharing, and appointment scheduling were developed with real use cases in mind, often shaped by feedback from potential users.

We've also learned a lot about user experience, state management,

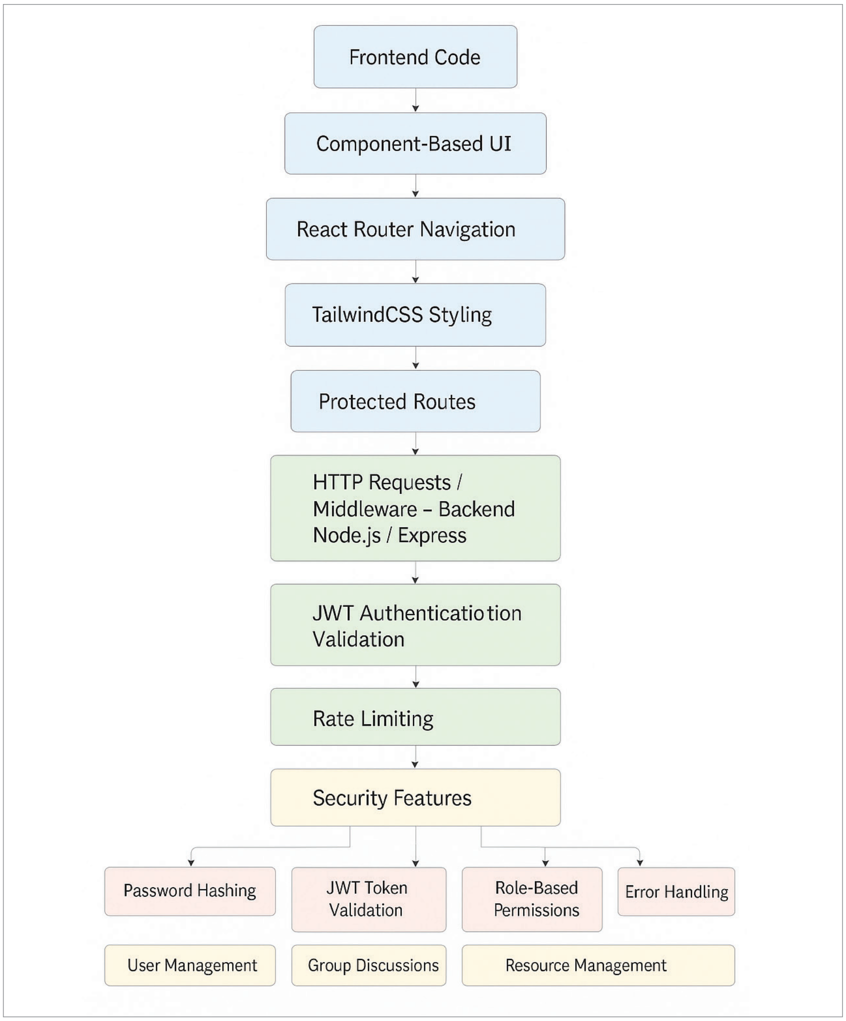

Figure 6: Frontend-backend integration diagram

security practices, and even the quirks of handling WebSocket connections in production. The project has grown beyond its original scope, and it continues to evolve as we explore new features like video consultations and ML-powered insights.

More importantly, SereneCare is open source. We hope that others will take this foundation and extend it in new directions, whether that's localising it for a community health programme, adapting it for a campus wellness centre, or simply learning from the architecture to build something better. END

Source code

The complete source code for the SereneCare platform is available as an open source project. Readers and developers are welcome to explore, fork, and contribute via the following GitHub repository: *https://github.com/AryanNukala/healthcare/tree/main*

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